Deliverable 1

CHU Decision-Support Data Repository

Conceptual Model · Architecture · Load Jobs

Field	Details
Institution	CESI
Group	Group 3
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1. Context and Objectives

The CHU (Cloud Healthcare Unit) group is pursuing digital transformation to leverage data from hospital management systems, flat files (satisfaction, deaths), and facility reference data.

Deliverable 1 establishes the decision-support data repository through three deliverables:

- Conceptual model of sources and star schema (OLAP).
- Big Data architecture description (ingestion → staging → DWH → BI).
- Integration job specifications and scripts (Talend + Hive/PostgreSQL).

2. User Requirements Analysis

2.1 Analysis Requirements

The table below lists the eight business requirements identified during the analysis phase.

ID	User Requirement	Output Type	Example (dummy data)
B1	Consultation rate at facility X over period Y	Time series / KPI	CH Fleyriat: 12,450 consultations in 2019
B2	Consultation rate by diagnosis X over period Y	Cross-tab diagnosis × month	Diagnosis J06: 320 consultations in Q1 2020
B3	Overall hospitalization rate over period Y	Aggregated KPI	2,479 stays over the covered period
B4	Hospitalizations by diagnosis over period Y	Histogram / table	Code S02800: 45 stays in 2017
B5	Hospitalizations by gender and age group	Stacked bar chart	Women 65+: 38% of stays
B6	Consultation rate per healthcare professional	Ranking / bar chart	Dr Martin: 890 consultations / year
B7	Number of deaths by region in 2019	Map or bars by region	Ile-de-France: 98,420 deaths (2019)
B8	Overall satisfaction by region (2020 campaign)	Average score by region	Nouvelle-Aquitaine: score 72.4 / 100

2.2 Requirements ↔ Analysis Areas

Requirement	OLAP Area	Fact Table
B1, B2, B6	Consultation	FACT_CONSULTATION
B3, B4, B5	Hospitalization	FACT_HOSPITALIZATION
B7	Death	FACT_DEATH
B8	Satisfaction	FACT_SATISFACTION

3. Operational Data Inventory (DATA 2024)

3.1 Source Summary

Source	Location	Format	Volume	Sensitivity
Deaths France	DECES EN FRANCE/deces.csv	CSV (,)	~25.1M rows	High (civil registry)
Hospitalizations	Hospitalisation/Hospitalisations.csv	CSV (;)	~2,479 rows	High
Healthcare facilities	Etablissement de SANTE/etablissement_sante.csv	CSV (;)	~416,665 rows	Low
Professionals	professionnel_sante.csv	CSV (;)	~1,048,576 rows	Medium
Professional activity	activite_professionnel_sante.csv	CSV (;)	~1,823,589 rows	Medium
Satisfaction (2014–2019)	Satisfaction/**	CSV (;)	Variable (~3,393 facilities IQSS 2019)	Low
Patient care paths	PostgreSQL (course install)	Relational	Consultations, patients, diagnoses	High

3.2 Key File Structures

Hospitalisations.csv

```
Num_Hospitalisation; Id_patient; identifiant_organisation; Code_diagnostic;
Suite_diagnostic_consultation; Date_Entree; Jour_Hospitalisation
```

deces.csv

```
nom; prenom; sexe; date_naissance; code_lieu_naissance; lieu_naissance;
pays_naissance; date_deces; code_lieu_deces; numero_acte_deces
```

sexe: 1 = male, 2 = female (civil registry).

etablissement_sante.csv

```
identifiant_organisation; finess_site; raison_sociale_site; commune;
code_commune; code_postal; ...
```

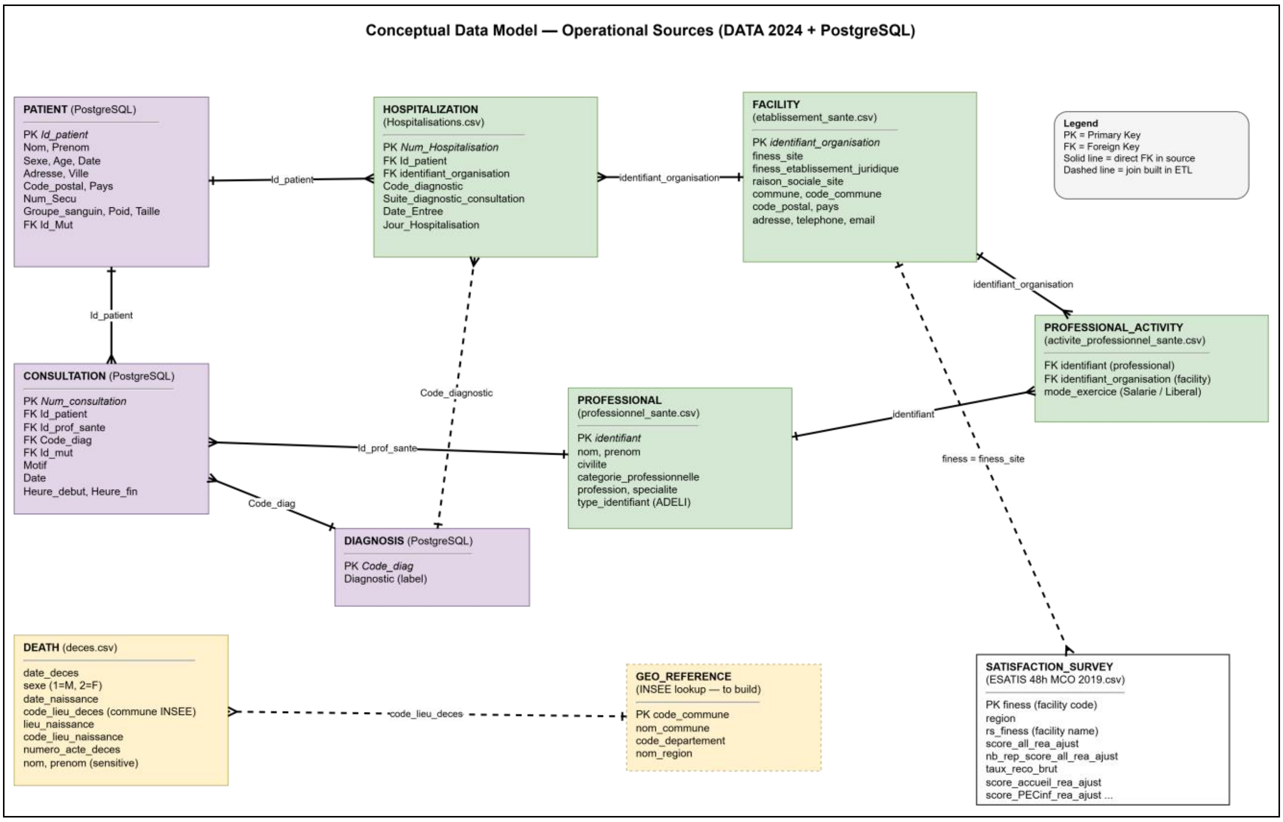
Satisfaction — ESATIS 48h MCO 2019

```
finess; region; score_all_rea_ajust; nb_rep_score_all_rea_ajust; taux_reco_brut; ...
```

PostgreSQL — Consultation

```
Num_consultation, Id_patient, Id_prof_sante, Code_diag, Motif, Date,
Heure_debut, Heure_fin, Id_mut
```

4. Conceptual Models by Source



4.1 PostgreSQL — PATIENT

Attribute	Role
Id_patient	PK
Nom, Prenom, Sexe, Age, Date	Demographics
Adresse, Ville, Code_postal, Pays	Location
Num_Secu, Groupe_sanguin, Poid, Taille	Medical admin
Id_Mut	FK (insurance)

4.2 PostgreSQL — CONSULTATION

Attribute	Role
Num_consultation	PK
Id_patient	FK → PATIENT
Id_prof_sante	FK → PROFESSIONAL
Code_diag	FK → DIAGNOSIS

Attribute	Role
Motif, Date, Heure_debut, Heure_fin	Visit details

4.3 PostgreSQL — DIAGNOSIS

Attribute	Role
Code_diag	PK (ICD-10)
Diagnostic	Label

4.4 CSV — HOSPITALIZATION

Attribute	Role
Num_Hospitalisation	PK
Id_patient	FK → PATIENT
identifiant_organisation	FK → FACILITY
Code_diagnostic, Suite_diagnostic_consultation	Diagnosis
Date_Entree, Jour_Hospitalisation	Stay duration

4.5 CSV — FACILITY

Attribute	Role
identifiant_organisation	PK
finess_site, raison_sociale_site	Identifiers / name
commune, code_commune, code_postal	Geography

4.6 CSV — PROFESSIONAL + ACTIVITY

PROFESSIONAL (professionnel_sante.csv): PK identifiant, nom, prenom, profession, specialite.

PROFESSIONAL_ACTIVITY (activite_professionnel_sante.csv): FK identifiant + FK identifiant_organisation, mode_exercice.

→ Links CONSULTATION to FACILITY when no direct facility key exists on the consultation row.

4.7 CSV — DEATH

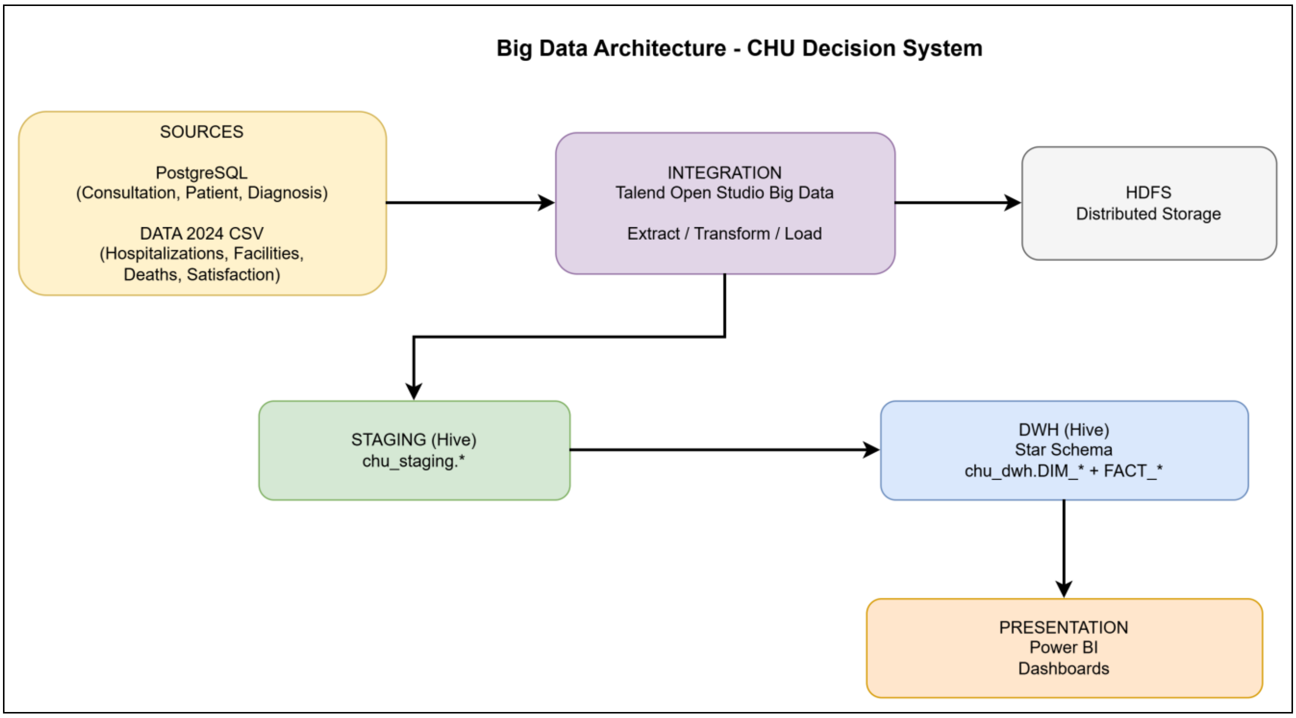
Attribute	Role
date_deces, sexe, date_naissance	Event + demographics
code_lieu_deces	FK → GEO_REFERENCE (commune)
numero_acte_deces	Registry id

Region is not in the file — built via GEO_REFERENCE (INSEE) in ETL.

4.8 CSV — SATISFACTION (ESATIS 48h MCO 2019)

Attribute	Role
finess	PK / FK → FACILITY.finess_site
region	Direct region column
score_all_rea_ajust, taux_reco_brut	Satisfaction KPIs
nb_rep_score_all_rea_ajust	Response count

5. Big Data Architecture for the Decision System



5.1 Layered Architecture

Layer	Description
PRESENTATION	Power BI — Dashboards & storytelling
DWH	Hive (chu_dwh schema) — Conformed dimensions + 4 fact tables (star schema)
STAGING	Hive (chu_staging schema) — Typed, cleansed, historized raw data
INTEGRATION	Talend Open Studio — Extract CSV / PostgreSQL → transform → load HDFS/Hive
SOURCES	DATA 2024 (CSV) + PostgreSQL + FTP (satisfaction/deaths)

5.2 Technical Components

Component	Role
PostgreSQL	Operational source for consultations / patients
HDFS	Distributed storage for staging and DWH
Hive	SQL engine on Hadoop — external and managed tables
Talend	ETL/ELT, flow orchestration
Power BI	Presentation (deliverable 3)

5.3 Data Flows

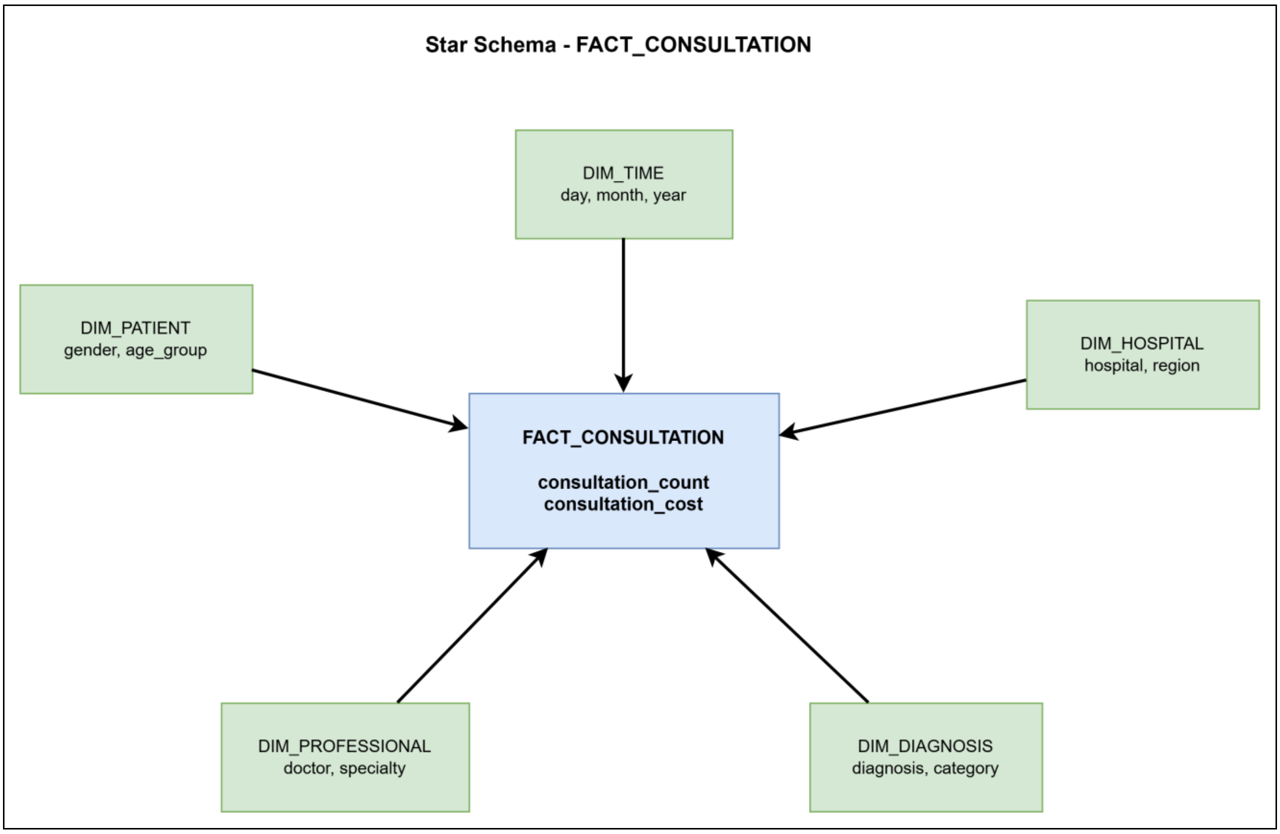
- Extract: read CSV (`;` or `,`) and PostgreSQL via JDBC.
- Transform: normalize dates, diagnosis codes, compute age_group, facility joins, municipality → region mapping (deaths).
- Load: chu_staging.* then chu_dwh.DIM_* and chu_dwh.FACT_*.
- Consumption: Hive analytical queries or export to Power BI.

5.4 Design Rationale

Choice	Rationale
Layered architecture	Staging/DWH separation, replay, traceability
Hive for DWH	Death volume (~25M rows), analytics, partitioning
4 star schemas	One business domain per fact, Power BI simplicity
Conformed dimensions	Consistent time / hospital / patient filters across facts
Talend	Course standard — PostgreSQL and flat-file connectors

6. OLAP Decision Model — 4 Analysis Areas

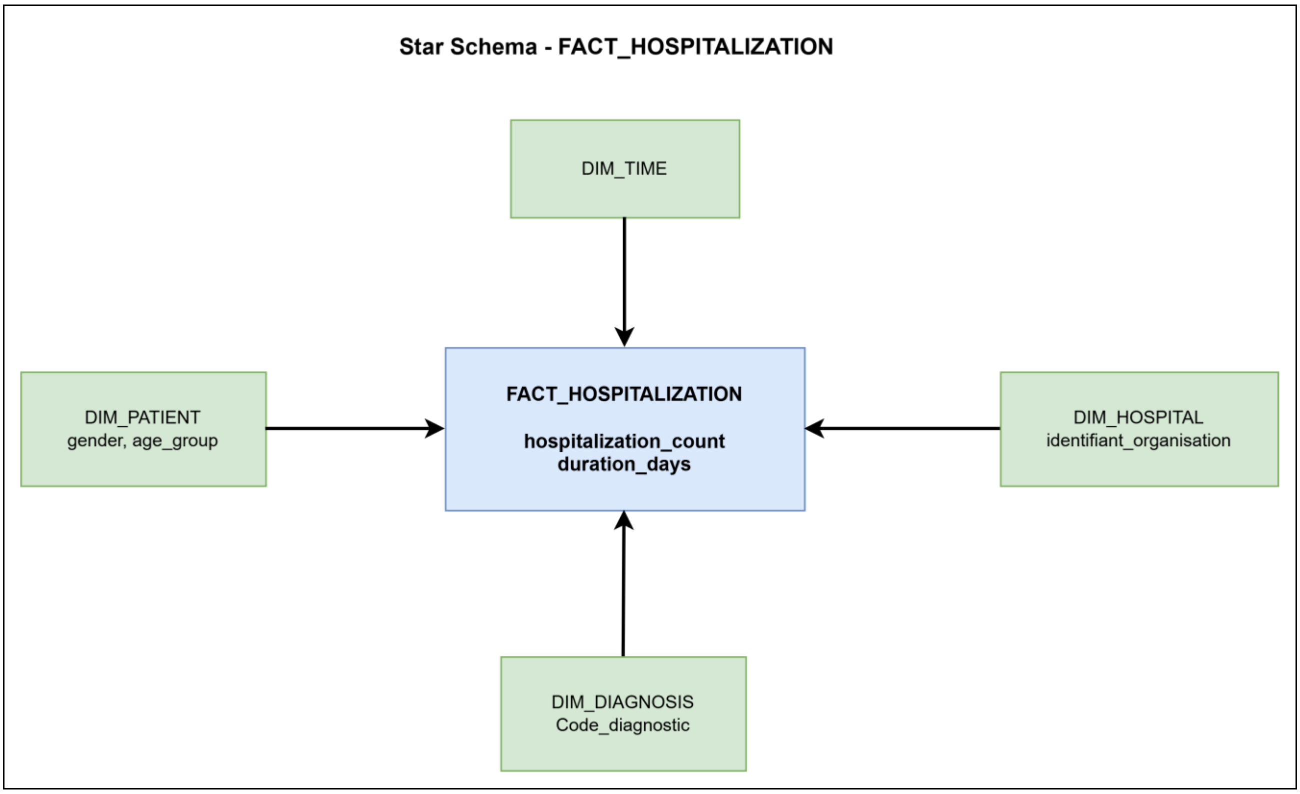
6.1 Area 1 — Consultations



Grain: 1 row = 1 consultation (Num_consultation).

Measure	Description	Aggregation
consultation_count	Counter	SUM
consultation_cost	Cost (if available later)	SUM

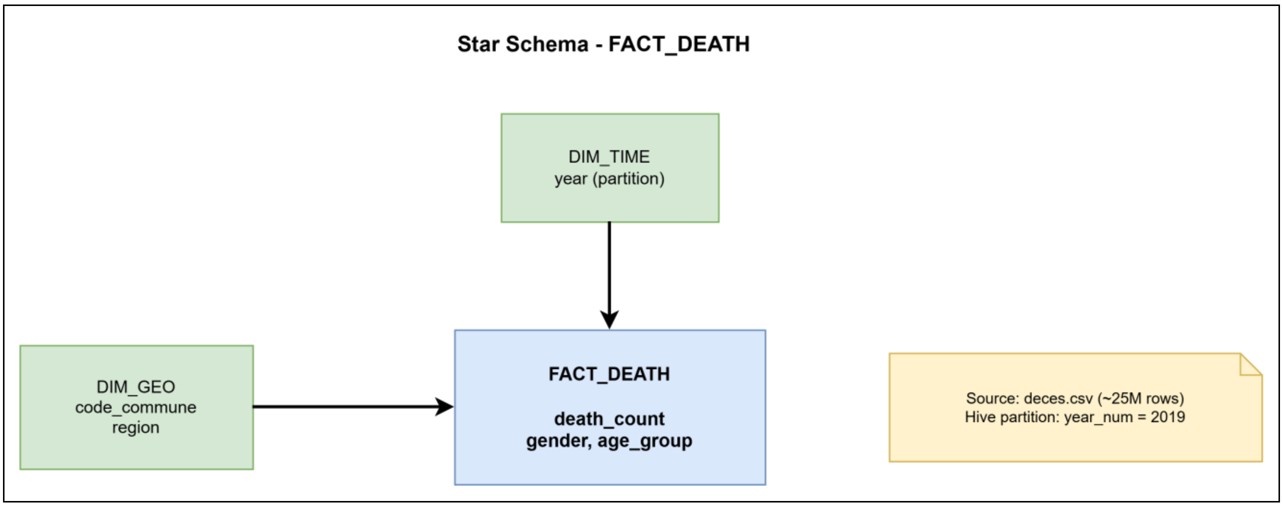
6.2 Area 2 — Hospitalizations



Grain: 1 row = 1 stay (Num_Hospitalisation).

Measure	Description
hospitalization_count	1 per stay
duration_days	Jour_Hospitalisation

6.3 Area 3 — Deaths

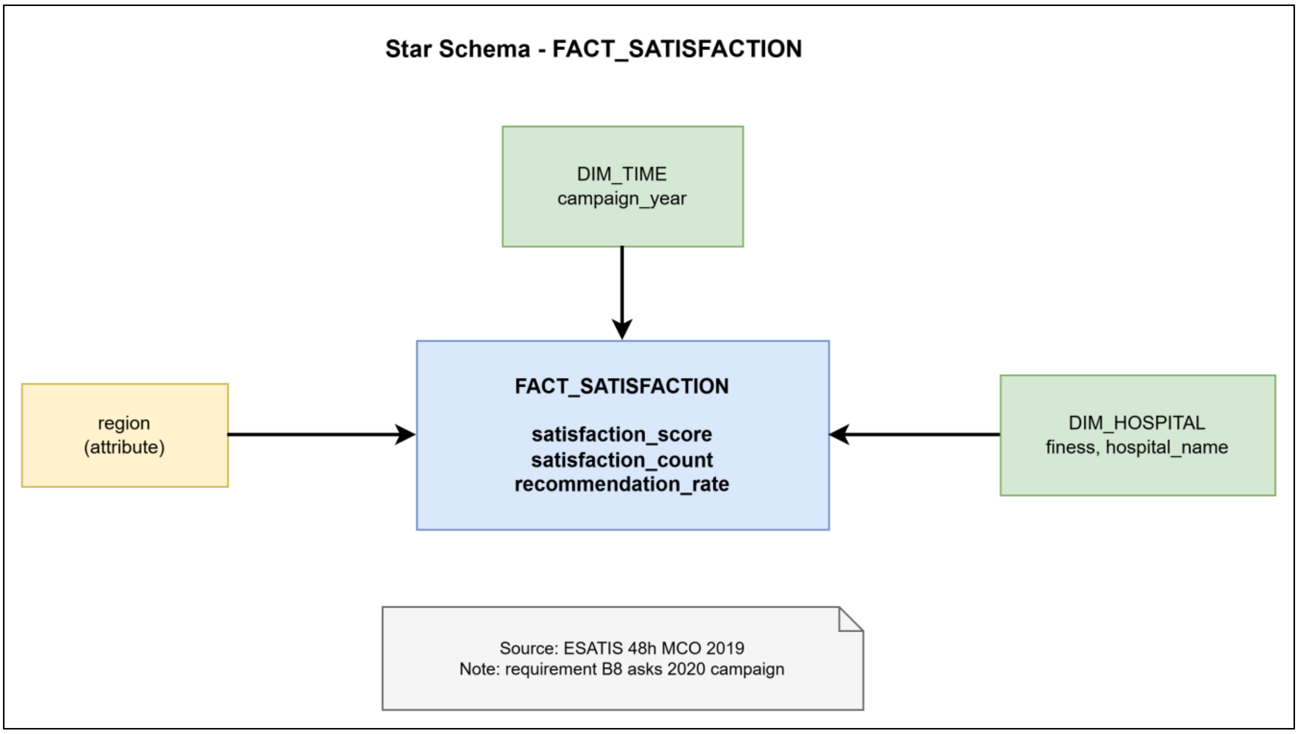


Grain: 1 row = 1 death record.

Measure	Description
death_count	1 per record

Recommended partitioning: year(date_deces) for Hive performance.

6.4 Area 4 — Satisfaction



Grain: 1 row = 1 facility × campaign × main indicator (ESATIS global score).

Measure	Description
satisfaction_score	score_all_rea_ajust
satisfaction_count	nb_rep_score_all_rea_ajust
recommendation_rate	taux_reco_brut

7. Conformed Dimensions and Measures

7.1 DIM_TIME

Attribute	Type	Source
time_sk	INT	Surrogate key
full_date	DATE	Event dates
day, month, quarter, year	INT	Derived
month_name	STRING	Derived

7.2 DIM_PATIENT

Attribute	Type	Source
patient_sk	INT	Surrogate key
patient_id	INT	Patient.Id_patient
gender	STRING	Patient.Sexe
age	INT	Patient.Age or computed from Date
age_group	STRING	0–17, 18–64, 65+

7.3 DIM_HOSPITAL

Attribute	Type	Source
hospital_sk	INT	Surrogate key
identifiant_organisation	STRING	etablissement_sante
finess_site	STRING	finess_site
hospital_name	STRING	raison_sociale_site
commune, code_commune	STRING	etablissement_sante
region	STRING	Satisfaction join or reference

7.4 DIM_DIAGNOSIS

Attribute	Type	Source
diagnosis_sk	INT	Surrogate key
code_diag	STRING	ICD-10
diagnosis_label	STRING	Label
category	STRING	ICD chapter (ETL grouping)

7.5 DIM_PROFESSIONAL

Attribute	Type	Source
professional_sk	INT	Surrogate key
professional_id	STRING	professionnel_sante.identifiant
last_name, first_name	STRING	nom, prenom
profession, specialty	STRING	profession, specialite

7.6 DIM_GEO (Deaths)

Attribute	Type	Source
geo_sk	INT	Surrogate key
code_commune	STRING	code_lieu_deces
commune_label	STRING	Reference
department	STRING	INSEE
region	STRING	Lookup table

8. Integration Jobs and DWH Loading

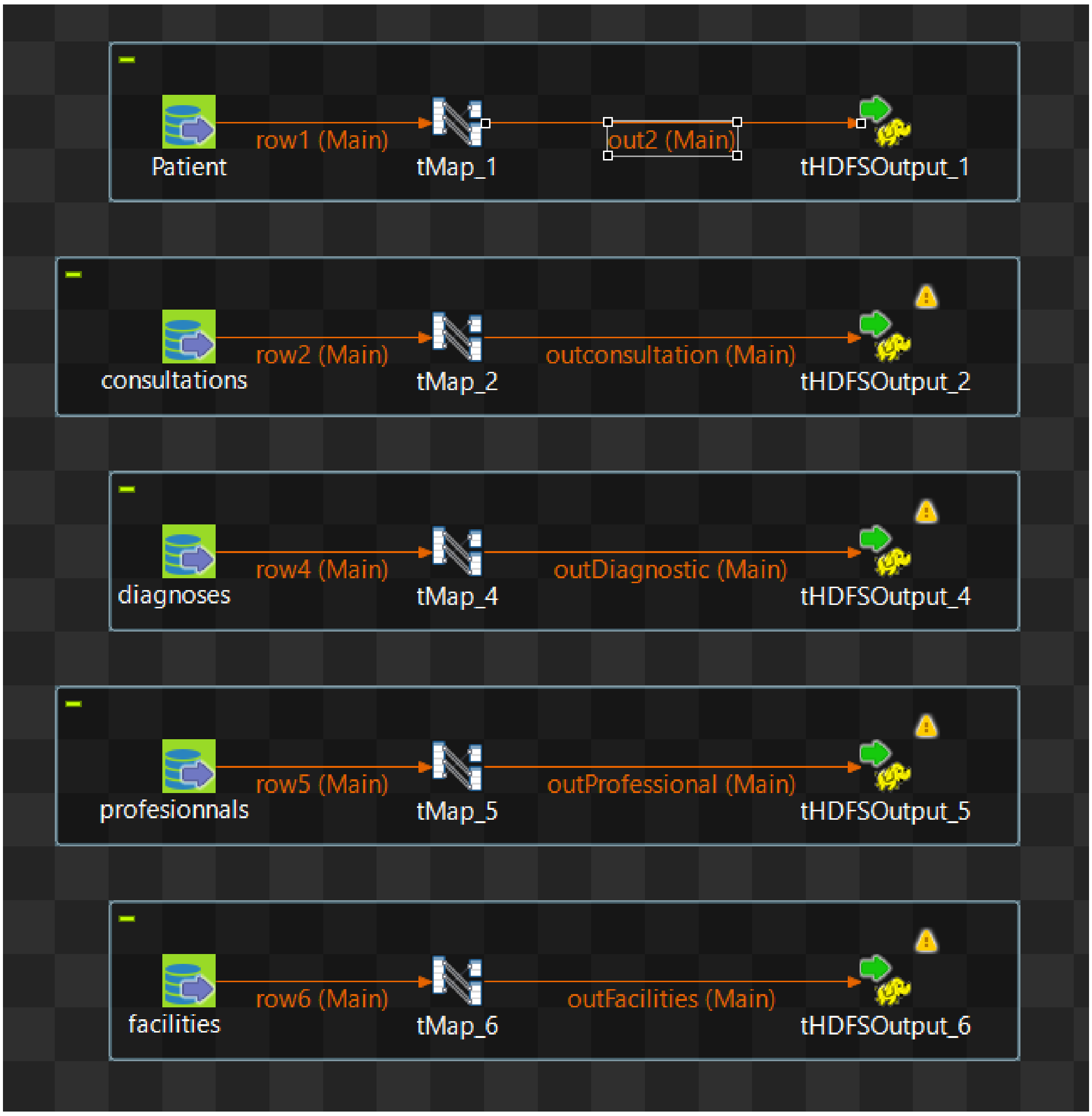
8.1 Job Summary

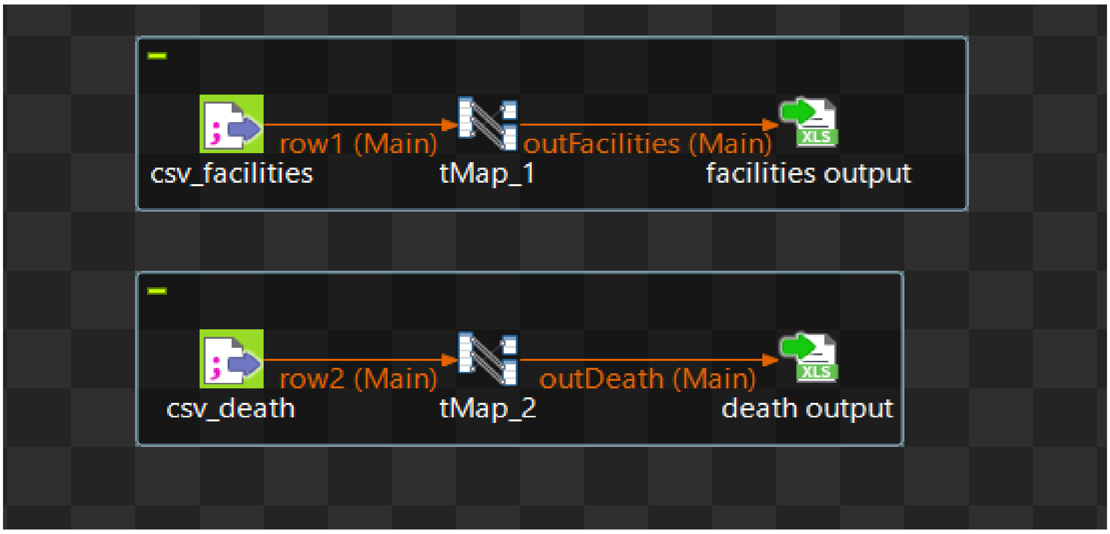
Job	Source	Staging Target	DWH Target
job_connexion_postgres	PosgreSQL (Patient, Consultation, Hospitalization, Diagnoses, Professionals, Facilities tables)	stg_patient, stg_consultation, stg_hospitalization , stg_diagnostic, stg_professional, stg_activity, stg_facility	DIM_PATIENT, FACT_CONSULTATION, FACT_HOSPITALIZATION, DIM_DIAGNOSTIC, DIM_PROFESSIONAL, DIM_HOSPITAL
job_csv	CSV (etablissement_sante.csv, deces.csv (year partition))	stg_facility, stg_death	DIM_HOSPITAL, FACT_DEATH
job_flat_files	Flat file (ESATIS 2019 (patient satisfaction survey))	stg_satisfaction	FACT_SATISFACTION
job_ftp	FTP (Death registry files: deces.csv)	stg_death	FACT_DEATH

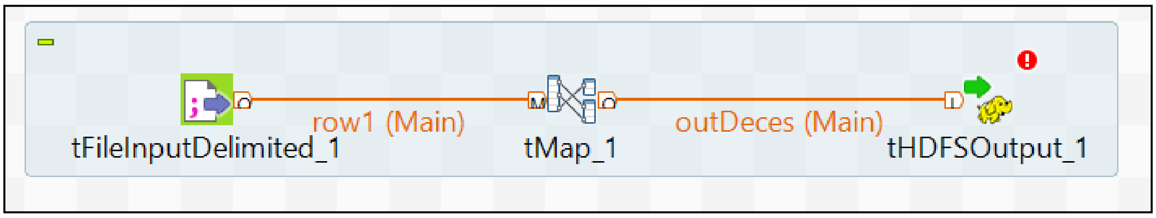
8.2 Main Transformation Rules

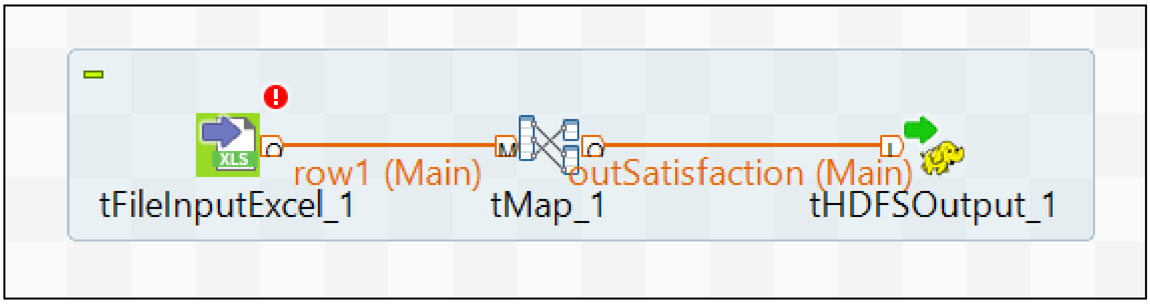
Rule	Description
Hospitalization dates	dd/MM/yyyy → yyyy-MM-dd
Patient age	Computed at event date for hospitalizations
Consultation ↔ facility	Consultation.Id_prof_sante → activite_professionnel_sante → identifiant_organisation
Death ↔ region	Join code_lieu_deces on municipality reference (INSEE file or derived table)
Data quality	Reject rows without mandatory keys; log rejects in chu_staging.rejects_*

8.4 Talend Screenshots









9. Security, Governance, and Data Volumes

Risk	Mitigation
Health / civil registry data	Restricted HDFS/Hive access; no name/surname in analytical DWH (deaths: aggregates only in BI)
25M death rows	Hive partition by year; no SELECT * in production
Traceability	Timestamped staging tables; Talend logs
GDPR	Minimization: technical IDs and aggregated dimensions for reporting

10. Conclusion and Outlook for Deliverable 2

Deliverable 1 establishes the CHU decision system foundations across three pillars:

- 4 analysis areas as star schemas covering all 8 user requirements.
- Layered architecture aligned with the course Big Data stack.
- Jobs and Hive SQL scripts ready for physical implementation.

Deliverable 2 will focus on the following steps to do in our project:

- Physical implementation of Hive tables and load scripts.
- Partitioning and bucketing strategies for performance.
- Response-time benchmark charts and validation.

11. Appendices

The following artefacts are provided alongside this report:

- annexes/mapping_requirements_sources.csv
- sql/*.sql — Hive scripts
- diagrams/*.drawio — Conceptual and star-schema models
- jobs/talend_job_specifications.md